

Day 04 â Paperclip 100% Closure

Date: 2026-05-09

Branch: to-knowledge-mc

Status: DEVELOPER-SIDE CLOSED

Blocker classification: SERVICE_DOWN

Executive Summary

Day 04 closed the developer-side Paperclip gate with truthful runtime proof. Paperclip is wired as the Workforce Control Plane before OpenClaw+, but the sandbox service is not currently running on the expected local/Tailnet endpoint. The status route now returns a canonical closure status and proof packet instead of a vague degraded state.

Paperclip is not marked GO. The current blocker is:

- 'paperclip_sandbox_service_not_running'

No task creation, workforce mutation, external write, or fake Paperclip success was enabled.

What Changed

- Added canonical Paperclip closure fields to the read-only status payload:
 - 'canonical_status'
 - 'blocker_class'
 - 'proof_packet'
- Added 'buildPaperclipClosureSummary()' for deterministic closure classification.
- Classified unreachable Paperclip sandbox runtime as:
 - 'canonical_status=SERVICE_DOWN'
 - 'blocker_class=SERVICE_DOWN'
 - 'blocker=paperclip_sandbox_service_not_running'
- Preserved existing Paperclip safety posture:
 - local or Tailnet endpoint only
 - no public exposure
 - execution disabled
 - writes disabled
 - protected actions disabled
 - Bridge Session required for mutations
- Added tests for Paperclip closure classification and proof packet safety.

Files Changed

- 'src/lib/paperclip-bridge.ts'
- 'src/lib/paperclip-bridge.test.ts'

Routes / Endpoints Changed

- 'GET /api/bridge/paperclip/status'
 - Returns canonical closure status and proof packet.
 - Continues to require authentication.
 - Continues to be read-only.

Read-only inventory routes remain unchanged in behavior, but were smoke-tested:

- 'GET /api/bridge/paperclip/companies'
- 'GET /api/bridge/paperclip/agents'
- 'GET /api/bridge/paperclip/issues'

Runtime / Service Proof

Local Paperclip service checks:

- 'ls -nlp -iTCP:3100 -sTCP:LISTEN': no listener found.
- 'curl http://127.0.0.1:3100/api/health': connection refused.
- 'systemctl --user status paperclip.service': no usable service status returned.
- Candidate path '/Users/sosastrike/Documents/New project/paperclip' is a zero-byte file, not a startable Paperclip repo.

Standalone Mission Control smoke was run bound to '127.0.0.1:3337' only.

Unauthenticated proof:

- '/login' returned '200'.
- 'GET /api/bridge/paperclip/status' returned '401'.

Authenticated proof:

- 'GET /api/bridge/paperclip/status' returned '200'.
- Payload included:
 - 'ok=true'
 - 'mode=paperclip_status_read_only'
 - 'health=degraded'
 - 'canonical_status=SERVICE_DOWN'
 - 'blocker_class=SERVICE_DOWN'
 - 'reachable=false'
 - 'configured=false'
 - 'blocker=paperclip_sandbox_service_not_running'
 - 'proof_packet.result=SERVICE_DOWN'
 - 'proof_packet.blocker=paperclip_sandbox_service_not_running'
 - 'execution_enabled=false'
 - 'writes_enabled=false'
 - 'no_secrets_exposed=true'
 - 'raw_paths_exposed=false'

Authenticated inventory proof:

- 'GET /api/bridge/paperclip/companies' returned '503' with 'paperclip_sandbox_service_not_running'.
- 'GET /api/bridge/paperclip/agents' returned '503' with 'paperclip_sandbox_service_not_running'
- 'GET /api/bridge/paperclip/issues' returned '503' with 'paperclip_sandbox_service_not_running'
- All inventory responses stayed read-only and secret-safe.

UI Behavior

No UI layout was changed in this day.

Owner-facing truth for Paperclip remains:

- Paperclip is the Workforce Control Plane before OpenClaw+.
- Paperclip service health is not proven because the sandbox service is down.
- Mutating controls must stay Bridge-gated or disabled with the exact blocker.

Tests And Validation

- 'git diff --check': PASS
- 'pnpm run typecheck': PASS
- 'pnpm run build': PASS
- 'pnpm test src/lib/paperclip-bridge.test.ts src/lib/paperclip-bridge-routes.test.ts': PASS, 42 tests
- 'pnpm test': PASS, 138 files / 1257 tests
- 'node scripts/check-protected-file-invariants.mjs': PASS
- staged secret scan: PASS
- '.env' diff check: clean

Safety Confirmation

- No '.env' edits.
- No secrets printed.
- No auth weakening.
- No public local exposure.
- No Paperclip write adapter enabled.
- No task creation.
- No external writes.
- No fake Done.
- No fake Paperclip health.
- No raw path exposure in owner-facing payloads.
- Agent Zero remains commander.
- Paperclip remains before OpenClaw+ in the operating chain.

Blockers

Current blocker:

- 'paperclip_sandbox_service_not_running'

Blocker class:

- 'SERVICE_DOWN'

Required owner/admin or runtime action:

1. Install or restore the Paperclip sandbox service/repo.
2. Ensure it listens only on loopback or Tailnet.
3. Provide '/api/health'.
4. Restart Mission Control only if endpoint configuration changes.
5. Re-run:
 - 'GET /api/bridge/paperclip/status'
 - 'GET /api/bridge/paperclip/companies'
 - 'GET /api/bridge/paperclip/agents'
 - 'GET /api/bridge/paperclip/issues'

Commit / Push

Code commit:

- 'bc1f9c1'
- Full hash: 'bc1f9c1dc2d9d184f5dfd84824d124953a8c2b3c'
- Message: 'feat(paperclip): add closure status proof packet'
- Push: 'origin/to-knowledge-mc' updated successfully.

Rollback:

```
""bash
git revert bc1f9c1
""
```

Next Day Started

Day 05 â OpenClaw+ 100% Closure has started automatically.